MB Note

1 HIV|SIR Model

HIV: $\begin{cases}\dot{x} = \lambda - dx - \beta xv \\ \dot{y} = \beta xv - ay \\ \dot{v} = ky - uv\end{cases} \Rightarrow$
- λ: formation rate of healthy cells.
- d: death rate of healthy cells.
- β: infection rate.
- a: death rate of infected cells.
- k: formation rate of free viruses.
- u: death rate of free viruses.

$\dot{v}$ 中缺 $-\beta xv$ 的原因:: The model neglects the cost of virus when it infects healthy cells.

virus-free steady-state: $y = 0$ and $v = 0$, possible $x = \frac{\lambda}{d}$.

Extimate virus increase: ***in each virus generation|over the characteristic lifetime*:**

Have: $(x, y, v)(0) = (\frac{\lambda}{d}, 0, v_0)$. **1.** Period of free virus lifetime: u^{-1}. Period of infected cell lifetime: a^{-1}.

2. $\frac{1}{u} \ll \frac{1}{a}$ **3.** $v(\frac{1}{u}) - v(0) = \int_0^{\frac{1}{u}} \dot{v}\,dt = k\int_0^{\frac{1}{u}} y\,dt - u\int_0^{\frac{1}{u}} v\,dt$

a1. $\int_0^{\frac{1}{u}} y\,dt \simeq \int_0^{\frac{1}{u}} y(\frac{1}{a})dt$ **b.** $\int_0^{\frac{1}{u}} v\,dt \simeq \int_0^{\frac{1}{u}} v(0)dt$, **a2.** $y(\frac{1}{a}) = \int_0^{\frac{1}{a}} \dot{y}\,dt \simeq \int_0^{\frac{1}{a}} \dot{y}(0)dt$

Basic Reproductive ratio: $R_0 = \frac{k\beta\lambda}{adu}$. $R_0 > 1$ virus invades; $R_0 < 1$ virus dies out.

估计 (e.g. R_0) 某个值范围 | 最后趋近于某个值 | 得到趋近速率: **1.** 计算 equilibrium. **2.** 计算 Jacobian 算 eigenvalues. **3.** 取最大 eigenvalue $\lambda \to$ 得 sol $\propto e^{\lambda t}$. **4.** 根据题意估计要求值范围 | 趋近值 | 速率 (λ).

(e.g.) 证明 $u =$ cont. when $t = \mathcal{O}(u^{-1})$: **1.** Linearise 估算趋近值 **2.** 假设原式 $\simeq$ 估计值 **3.** 解出 t 估计

SIR: $\begin{cases}S' = -\beta SI \\ I' = \beta SI - \alpha I \\ R' = \alpha I\end{cases} \Rightarrow$
- S: susceptible individuals (易感人群).
- I: infected individuals (感染者).
- R: recovered individuals (康复者).
- β: infection rate.
- α: recovery rate.

ps: $\lim_{t\to\infty} I(t) = 0$ $\lim_{t\to\infty} S(t) > 0$ and $R(0) = 0$

Assumption: No births/deaths/migration.
Total Pop.: $N = S + I + R$ (constant).
Basic Reproductive Number: $R_0 = \frac{\beta N}{\alpha}$.
$R_0 > 1 \Leftrightarrow S > \frac{\alpha}{\beta}$ (epidemic occurs).
$R_0 < 1 \Leftrightarrow S < \frac{\alpha}{\beta}$ (disease dies out).

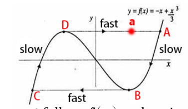
Final Size Relation: $\ln(S_0/S_\infty) = \frac{\beta}{\alpha}[N - S_\infty] = R_0(1 - \frac{S_\infty}{N})$

proof: **1.** $(S+I)' = -\alpha I$ **2.** $\int_0^\infty (S+I)'dt = -\alpha\int_0^\infty I\,dt \Rightarrow (S+I)_\infty - (S+I)_0 = \int_0^\infty \frac{\alpha S'}{\beta S}dt$

3. By $N = S_0 + I_0$, can get $S_\infty - N = \ln(S_\infty/S_0)\frac{\alpha}{\beta}$. ☐

2 Relaxation|Excitation|Switch - FitzHugh Nagumo Model

Explain why oscillations|relaxation occur:



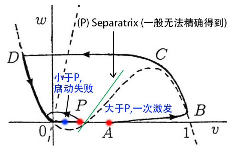
1. (e.g.) $\dot{x} = c[f(x) - y]$ and $\dot{y} =$ a line, where $f(x)$ is a cubic-like curve. $c \gg 1$
2. show $|\dot{x}| \gg |\dot{y}|$ (or vice versa), and no stable equilibrium exists. (can unstable)
3. 图上标一起始点 **a**, 画类似右侧图. **4.** (e.g.) **a.** Start at **a**, $|\dot{x}| \gg |\dot{y}|$, so it quickly moves horizontally to the right branch of cubic-like curve (A). **b.** When it closed to the right branch, since $\dot{y} < 0$, $\dot{x} \simeq 0$, so it moves downwards along the right branch. Then, it closes to the point B. It has option to leave the path $f(x)$ since $\dot{y} < 0$, so it cannot follow $f(x)$ and again quickly jumps from B to the left branch of the cubic-like curve (C) since $|\dot{x}| \gg |\dot{y}|$. When it closed to the left branch, since $\dot{y} > 0$, $\dot{x} \simeq 0$, so it moves upwards along the left branch. Finally, it reaches the point D, and again quickly jumps to the right branch of the cubic-like curve (A). This process repeats, resulting in relaxation oscillations.

Estimate Period: (neglect fast jumps) **1.** Period $T \simeq \int_{t_A}^{t_B} dt + \int_{t_C}^{t_D} dt = 2\int_{t_A}^{t_B} dt$ (by symmetry)

2. (e.g.) $\dot{x} = c[f(x) - y]$, $\dot{y} = g(x, y)$ then $y \simeq f(x)$, $\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt} = f'(x)\frac{dx}{dt} \Rightarrow dt = \frac{f'(x)}{g(x,y)}dx$

3. Find coordinates of A, B, C, D (turnning points) $\Rightarrow T \simeq 2\int_{x_A}^{x_B} \_\, dx$

Three kinds of Excitable System:



small perturbation: returns to equilibrium. (fail excitation)
large perturbation: large excursion in phase space before returning to equilibrium. (action potential|successful excitation)
Refactory period(不应期): system insensitive to further
constant stimulus: periodic excitation. (repetitive firing|limit cycle)
Explain why excitable occur: 结合前面 relaxation oscillation 的分析, stable equilibrium (图为原点), 分析 P|A 开始 (small|large perturbations).

FitzHugh-Nagumo Model:

Have: $\begin{cases}\dot{v} = f(v) - w + I(t) \\ \dot{w} = bv - \gamma w\end{cases} \Rightarrow$ v: membrane (excitable) potential. w: ionic current. (recovery var.) a,b,c: positive constants. I(t): external stimulus.

Switch Behaviour: 类似 relaxation oscillation, 有two stable equilibria, 分析一个过去后不会回来.

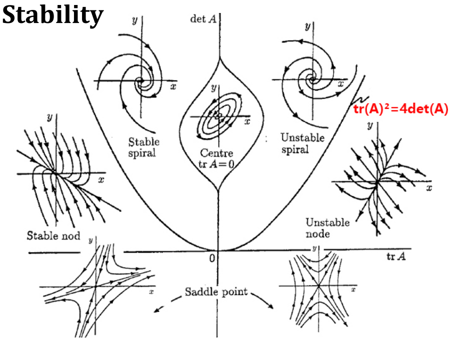
Remark: 对于这类模型, 计算平衡点用 $f(v)$ 的图像斜率和另一条直线斜率来判断.

3 Stability|Linearise Stability

Continuous $\frac{dN}{dt} = f(N)$:
- **Stable**: $f'(N^*) < 0$
- **Unstable**: $f'(N^*) > 0$

Discrete $N_{t+1} = f(N_t)$: $\lambda = f'(N^*)$
- **Stable**: $|\lambda| < 1$ - **periodic**: $|\lambda| = 1$
- **Unstable**: $|\lambda| > 1$ (chaotic)
- **tangent bifurcation**: $\lambda = 1$
- **period-doubling bifurcation**: $\lambda = -1$

One var. ↑ | **More var.** ↓:



Jacobian: $J = [\mathbf{F}_x\ \mathbf{F}_y\ \cdots]$
Eigenvalue Method:
If Re$(\lambda_i) < 0$ for all i, then asy.stable.
If $\lambda_1 < 0 < \lambda_2$, then saddle (unstable).
If Im$(\lambda_i) \neq 0$, then focus (spiral).
If Re$(\lambda_i) = 0$, then center (linear!).
or:(nolinear): **Hopf bifurcation.**

If $\lambda = 0$: If $\lambda_1 = 0$, $\lambda_2 < 0$: It's a "saddle-node" with one stable eigendirection (corresponding to $\lambda_2 < 0$) and one "center" eigendirection (corresponding to $\lambda_1 = 0$). ps: 如果 $\lambda > 0$ 或都 $= 0$, 类似分析.

Linearise Stability: **1.** Let $N(t) = N^* + n(t)$, $|n(t)| \ll 1$. or $N_t = N^* + n_t$, $|n_t| \ll 1$.

2. $\frac{dn}{dt} = \frac{dN}{dt} =$ **a.** $\overset{0}{\cancel{f(N^*)}} + f'(N^*)n(t)$ or $\overset{N^*}{\cancel{f(N^*)}} + \overset{\lambda}{\cancel{f'(N^*)}}n_t$ **b.** (delay) 直接代入后 linearise.

Justify bifurcation: **1.** Assume $n(t) = e^{\lambda t}$ **2.** 代入 $\frac{dn}{dt}$: $\lambda e^{\lambda t} = f'(N^*)e^{\lambda t}$ **3.** (Re), (Im) 得两方程.
4a. 反证 $\Re(\lambda) > 0, < 0$ or **4b.find T_c** Check $T = 0$ stable|unstable $\Rightarrow$ first time $\Re(\lambda) = 0$

Properties for continuous: **Time-Independent**: At N^*, $N(t - T) = N(t) = N^*$.
Tangency-Condition: At bifurcation point (x^*, r_c), $f_x(x^*, r_c) = f(x^*, r_c) = 0$.

Properties for discrete: **eigenvalue**: $\lambda = f'(N^*)$.
(for perturbations) $\lambda < -1$: ↑ oscillation; $-1 < \lambda < 0$: ↓ oscillation;
$0 < \lambda < 1$: ↓ monotonic; $\lambda > 1$: ↑ monotonic.
p-periodic: $N^* = f^p(N^*)$. Every p-periodic N^* with $\lambda = -1$ branches into a $2p$-periodic N^*.
Sarkovskii's Theorem: If a 3-periodic N^* exists, then p-periodic N^* exists for all $p \in \mathbb{N}^+$.
Max/Min: $f'(N_m) = 0$, $N_{\max} = f(N_m)$, $N_{\min} = f(N_{\max})$. assume *unimodal*, 2-periodic exists.

Types of Bifurcations: **Saddle-Node**: $\dot{x} = r \pm x^2$ (Normal Form) $0 \to$ S $\to$ U-S
Transcritical: $\dot{x} = rx - x^2$ (Normal Form) S-U $\to$ U-S
super-critical Pitchfork: $\dot{x} = rx - x^3$ (Normal Form) S $\to$ S-U-S
sub-critical Pitchfork: $\dot{x} = rx + x^3$ (Normal Form) U $\to$ U-S-U

Hopf: (平衡点从稳定变为不稳定, 产生 **limit cycle**) **Condition**: $\det(J) > 0$, $\frac{d\mathrm{tr}(J)}{d\mu} \neq 0$. μ: parameter, not t
or: 找到 μ s.t. tr$(J) > 0$, $\mu > \mu_c$ and tr$(J) < 0$, $\mu < \mu_c$. **Period**: $T = \frac{2\pi}{w}$, 特征向量: $\lambda = \pm iw$

Bifurcation Diagram: x-axis: equilibrium value; y-axis: parameter value. 虚线: unstable; 实线: stable

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 # ODEs

$y' + py = q$: Integrating Factor: $I = e^{\int p\,dt}$ sol: $I(x)y = \int I(x)q\,dt$

$y'' - \lambda y = 0$: **1.** $\lambda > 0$: $y = c_1\sinh(\sqrt{\lambda}t) + c_2\cosh(\sqrt{\lambda}t) = c_1' e^{\sqrt{\lambda}t} + c_2' e^{-\sqrt{\lambda}t}$
2. $\lambda = 0$: $y = c_1 t + c_2$ **3.** $\lambda < 0$: $y = c_1\sin(\sqrt{-\lambda}t) + c_2\cos(\sqrt{-\lambda}t)$

$ay'' + by' + cy = 0$: **1.** $\Delta > 0$: $y = c_1 e^{r_1 t} + c_2 e^{r_2 t}$ $r_{1,2} = \mu + vi$
2. $\Delta = 0$: $y = e^{rt}(c_1 + c_2 t)$ **3.** $\Delta < 0$: $y = e^{\mu t}(c_1\cos vt + c_2\sin vt)$

4 Nullclines|Limit Cycle|Trapping Region

Nullclines: For: $u' = f(u, v)$, $v' = g(u, v)$. **u-nullcline**: $f(u, v) = 0$. **v-nullcline**: $g(u, v) = 0$.

Slope of Nullclines: Since $df = f_u du + f_v dv = 0$ on u-nullcline,
Slope of u-nullcline: $\frac{dv}{du} = -\frac{f_u}{f_v}$. Similarly, $\frac{dv}{du} = -\frac{g_u}{g_v}$ for v-nullcline.

Trapping Region: **1.** 直线/nullcline 围闭合区域. **2.** 边界上, 外法向量 $\mathbf{n}$ 有 $\mathbf{n} \cdot \begin{bmatrix} f(u,v) \\ g(u,v)\end{bmatrix} < 0$

Poincaré-Bendixson: One unstable equilibria (no saddle) + trapping region = stable limit cycle exists.

Useful Tools: $rr' = xx' + yy'$ and $r^2\theta' = xy' - yx'$.

5 Lotka-Volterra Model —Predator-Prey Model

coalesce: 两点合并 eradicated: 根除 membrane potential: 膜电位 unimodal: 单峰 vaccination: 疫苗接种
Predator: 捕食者 Prey: 被捕食者
boundary region = trapping region = invariant region = confined region